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Evaluation of the microflora of public surfaces

CH5200 - Food Microbiology
Experiment 2a



Group 2		
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1 General Introduction

Microorganisms are prevalent on various surfaces, with varying levels of contamination depending on location and surface properties. Such microflora may contain pathogenic strains, which may cause illness (Boyce et al., 2007). Hence, surface sampling can be done to ensure that the microbial load of public surfaces is within safe limits. In this experiment, microbiological sampling and analysis was performed on a public lounge seat that is used on a regular basis to evaluate its microbial load.

2 Materials and Methods

In this experiment, four sampling techniques were used: dipslide, swab, ATP, and direct contact plate methods. Sampling was conducted on four different surface categories—fingertip, body parts, a general surface, and a flat surface. For fingertip imprinting, the fingertips of each member's right hand (excluding the thumb) were directly imprinted onto individual PCA agar plates. Three body parts (palm, neck, and forearm) and three general surfaces (laboratory floor, phone case, and iPad screen) were sampled using the dipslide method. As for the flat surface, a public lounge seat located within the school was sampled using the dipslide, swab, and ATP methods.

All samples were incubated at 30°C for 24 hours. The sample colonies were counted and recorded.

3 Results and Discussion

Table 1 contains the microbiological counts of the PCA agar plates that were imprinted with the fingertips of each of the three group members.

Table 1: Microbiological status of group members' fingertips

Members	Total CFU of 4 fingertips	Average CFU/ fingertip
1	37	9.25
2	88	22.00
3	167	41.75

As shown in Table 1, the average levels of CFU/fingertip of each member varied, with member 3 displaying a higher microorganism count. This variation in CFU count was expected. Due to a complex interaction between internal and external factors, each person's skin microbiome composition differs greatly (Khan & Koh, 2024). Differences in CFU counts amongst each member likely reflect differing surface contact prior to fingertip imprinting and varying personal hygiene habits.

Next, the dipslide method was used to sample various surfaces. Tables 2 and 3 contain the results interpreted using a model density chart (Annex Figure B1 and Table B2 compared to Table C1).

Table 2: Microbiological status of group members' body parts obtained via dipslide

Members	Body part	CFU/ 25cm ²
1	Forearm	25
2	Neck	125
3	Palm	25

Table 3: Microbiological status of general surfaces obtained via dipslide

Members	General surface	CFU/ 25cm ²
1	Ipad screen	25
2	Phone case	25
3	Laboratory floor	125

From Table 2, it can be inferred that the neck region harbored higher levels of microorganisms as compared to the wrist and palm regions. This may be attributed to the warmer and more humid environment at the neck region, which favours microbial growth. The neck is also less exposed to mechanical abrasion compared to the wrist and palm, which are more frequently cleaned or come into contact with surfaces that remove surface bacteria. Additionally, the neck is closer to mucosal surfaces, which may serve as microbial transfer sources.

Table 3 shows that the laboratory floor has a higher CFU count than the surfaces of the iPad and phone case. The floor is subjected to constant foot traffic, introducing and redistributing microorganisms from various sources. Furthermore, floors frequently gather organic matter, dust, and dirt, providing nutrients for bacterial growth. While personal devices are regularly cleaned due to their visibility and frequent handling, floors may not receive the same level of attention, over time allowing microbial growth and spread. Porous or textured floor surfaces may also harbor bacteria more effectively than smooth, non-porous surfaces of electronic devices.

Thereafter, the public lounge seat located within the school was sampled using 3 methods, with the results summarized in Table 4. Hygiene scores were also assigned according to a standard used by food factories (Annex Figure B1 and Table B2 compared to Table D1).

Table 4: Microbiological status of lounge seat surface in SBS Lobby, NTU

Method	CFU/ 25cm ²	Hygiene score	Hygiene status
Dipslide	25	2	Satisfactory
Swab	25	2	Satisfactory
Method	RLU/ 25cm ²	Hygiene score	Hygiene status
ATP	1356	-	Fail

Both the dipslide and swab methods yielded a hygiene score of 2, reflecting that while basic cleanliness standards are met, more effort could be spared for improvement. On the other hand, the ATP test failed the cleanliness assessment, suggesting the presence of residual organic matter or microbial contamination that is undetectable through culture-based methods (i.e. non-culturable microorganisms). ATP testing measures adenosine triphosphate, an essential molecule found in all living cells. It provides a rapid indication of overall surface contamination. The ATP test's sensitivity to organic residues or non-culturable microorganisms signifies insufficient cleaning, which

may be the cause of the discrepancy between the ATP results and the culture-based techniques. This emphasizes how crucial it is to use a variety of hygiene assessment techniques in order to get a thorough grasp of surface cleanliness and to spot possible areas that need more stringent sanitation regulations.

The oversensitivity of the ATP test towards non-microbial matter may render it unreliable occasionally when detecting microbial contamination. This unreliability can be observed from the results of all the groups (Annex E1).

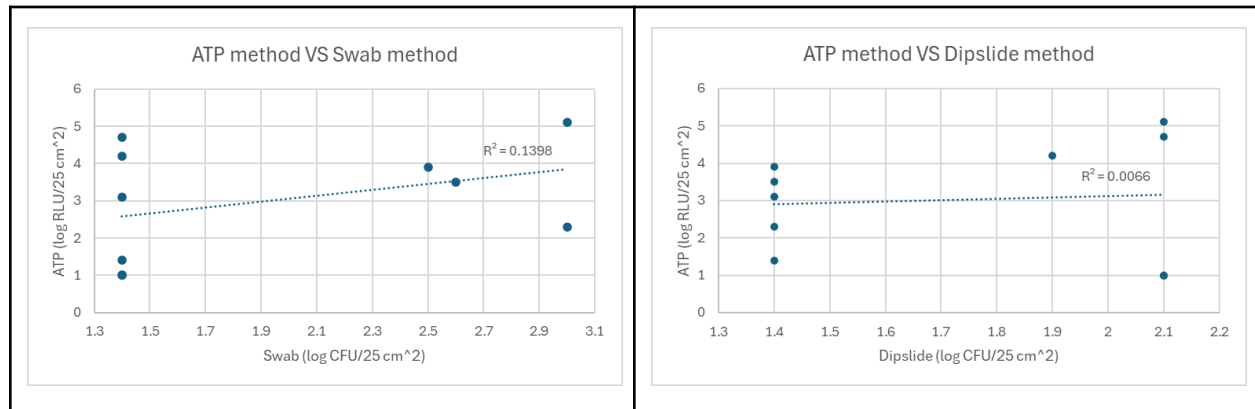


Figure 1: [left] ATP (log RLU/ 25cm²) against the Swab method (log CFU/ 25cm²)
[right] ATP (log RLU/ 25cm²) against the Dipslide method (log CFU/ 25cm²)

Based on Figure 1, the R^2 values (0.1398 and 0.0066) are close to 0, indicating little to no correlation between the dipslide/swab results and the ATP results, indicating that ATP might not be a suitable test for determining the microbiological status of a surface.

Question from the manual:

Is the ATP method suitable for quantitative measurements? Explain.

ATP (adenosine triphosphate) testing offers rapid quantitative assessments of surface cleanliness by detecting ATP levels via bioluminescence, with results expressed in Relative Light Units (RLUs). Although this approach gives relatively quick feedback on the existence of organic material, it is not very accurate at detecting microbial contamination. This restriction results from the fact that ATP tests identify all ATP sources, including non-microbial residues such as food particles or human cells, leading to overestimations of microbial presence. Studies have shown that there is a weak correlation between RLU values and aerobic colony count, with an R value of only 0.244 (Van Arkel et al., 2021). This study suggests that ATP may not be entirely reliable for assessing microbial contamination levels. Hence, while ATP testing provides prompt general cleanliness assessment, it should be complemented with culture-based methods to accurately evaluate and quantify microbial contamination.

4 Conclusions and Recommendations

This experiment assessed microbial contamination across various surfaces using direct contact plate, dipslide, swab, and ATP bioluminescence methods. Fingertip microbial variation among individuals reflected differences in the skin microbiome, recent contact, and hygiene habits. Culture-based methods revealed moderate microbial presence, with higher CFU counts on the neck and laboratory floor, which are areas that provide a more favorable environment for microbial growth due to exposure to organic matter, warmth, and moisture. While the lounge seat showed satisfactory results in culture-based assessments, it failed the ATP testing, highlighting the limitation of relying solely on ATP measurements for evaluating microbial contamination.

Based on the findings of these experiments, it is recommended to implement a multi-method approach to evaluate surface cleanliness, as no singular method could provide a fully comprehensive assessment. ATP testing rapidly provides results, but it is accompanied by the drawback of producing false positives due to its high sensitivity to all organic residues. Culture-based methods, which are better at reflecting viable microbial loads, should be implemented as a complement to more accurately assess microbial contamination.

Regular cleaning and sanitation, especially of high-contact or high-exposure areas like floors and shared seating, should be maintained or improved. Additionally, public awareness regarding hand hygiene remains crucial, as microorganisms are constantly present on human skin.

References

- Boyce, J. M. (2007). Environmental contamination makes an important contribution to hospital infection. *Journal of hospital infection*, 65, 50-54.
- Khan, A., & Koh, Y. (2024). The role of skin microbiome in human health and diseases. *Journal of Bacteriology and Virology*, 54(3), 191–202.
<https://doi.org/10.4167/jbv.2024.54.3.191>
- Van Arkel, A., Willemsen, I., & Kluytmans, J. (2021). The correlation between ATP measurement and microbial contamination of inanimate surfaces. *Research Square (Research Square)*. <https://doi.org/10.21203/rs.3.rs-428499/v1>

Annex

Table A1: Microbiological status (average CFU calculations) of group members' fingertips

Members	Total CFU of 4 fingertips	Average CFU/ fingertip
1	37	$\frac{37}{4} = 9.25$
2	88	$\frac{88}{4} = 22.00$
3	167	$\frac{167}{4} = 41.75$

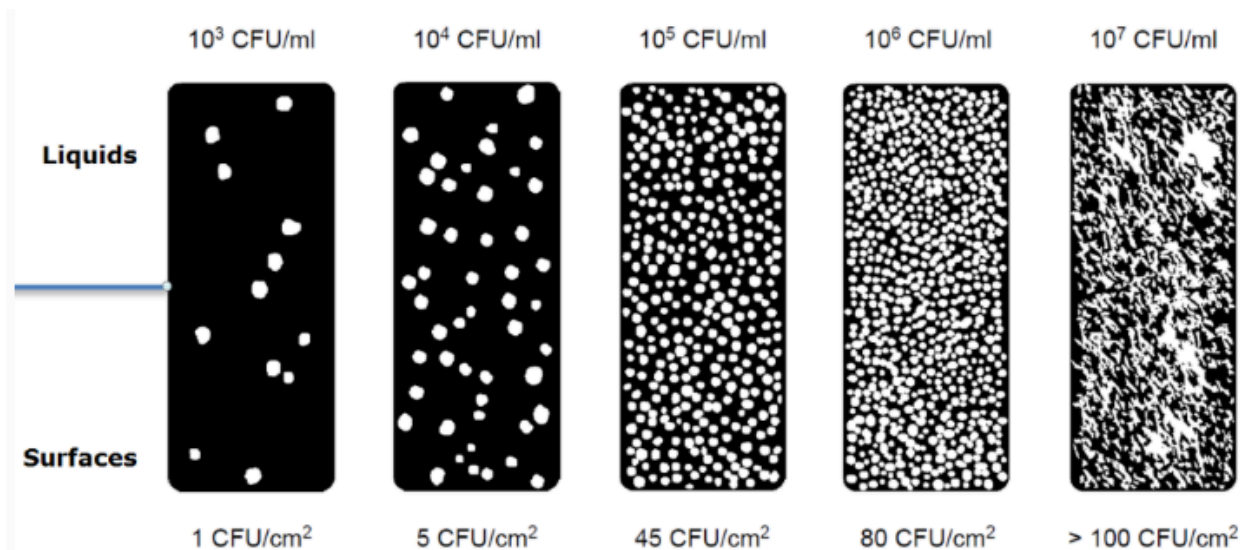


Figure B1: Model density chart used to determine microbial count, by comparing the microbial density on the dipslide with the chart

Table B2: Hygiene score standard for food factory

Number of CFU on 25cm ²	Score	Remark
0-2	0	Very Good
3-9	1	Good
10-29	2	Satisfactory
30-90	3	Questionable
>90	4	Unsatisfactory

Table C1: Microbiological status of group members' body parts and general surfaces obtained via dipslide method

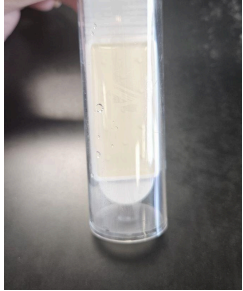
Members	Body part	Dipslide for body part	CFU/ 25 cm ²	General surface	Dipslide for general surface	CFU/ 25 cm ²
1	Forearm		25	Ipad screen		25
2	Neck		125	Phone case		25
3	Palm		25	Laboratory floor		125

Table D1: Microbiological status of lounge seat surface in SBS Lobby, NTU

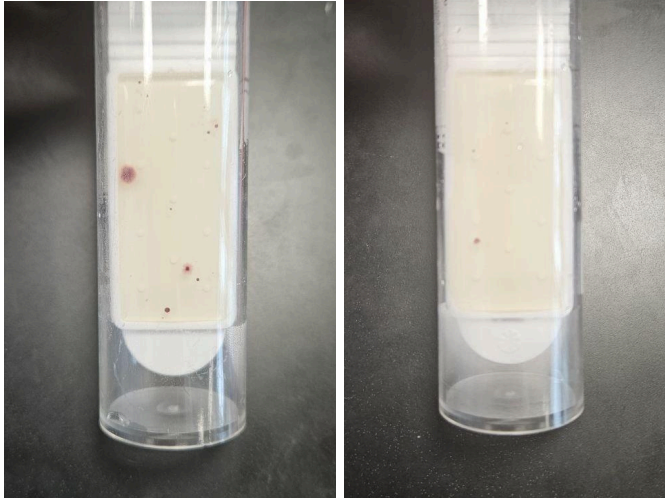
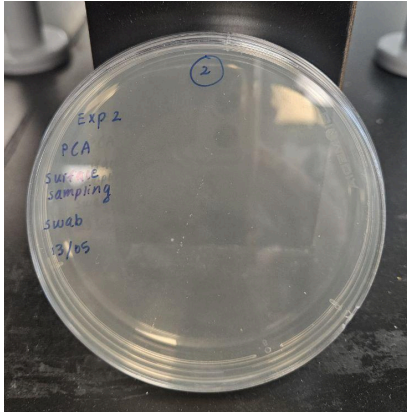
Method	Results	CFU/ 25cm ²
Via dipslide	 *Both sides of the dipslide were used for sampling	25
Via swab		25
Method	Results	RLU/ 25cm ²
Via ATP bioluminescence	-	1356

Table E1: Results for surface sampling from all groups

	CFU/25 cm ²		log CFU/25 cm ²		RLU/25 cm ² (Pass/ Fail)	log RLU/25 cm ²
Surface	Via dipslide	Via swab	Via dipslide	Via swab	Via ATP	Via ATP
Lab table	25	25	1.4	1.4	25 (pass)	1.4
SBS Lobby couch	25	25	1.4	1.4	1356 (fail)	3.1
Lift button	125	<25	2.1	<1.4*	0** (pass)	0
Water cooler drain	125	1120	2.1	3.0	129648 (fail)	5.1
Toilet seat	125	<25	2.1	<1.4*	0** (pass)	0
Laboratory sink 1	25 - 125 (75 was used)	<25	1.9	<1.4*	17291 (fail)	4.2
Toilet rim	25	350	1.4	2.5	8141 (fail)	3.9
Laboratory sink 2	<25	900	<1.4*	3.0	214 (fail)	2.3
Laboratory sink 3	<25	430	<1.4*	2.6	3396 (fail)	3.5
Toilet door handle	125	<25	2.1	<1.4*	49586 (fail)	4.7

*It is important to note that microbial counts of <1.4 log CFU/25 cm² were taken as 1.4 when plotting the graph.

**Value of 1 was used when calculating log RLU/25 cm² to prevent math error.